LoRA on a different task: SMS spam classification

Real-message classification rather than sports question answering.

Data: 1200 train / 400 validation / 771 test; three training seeds.

Duplicate/template groups stay in one split. First 96 GPT-2 message tokens; classify ham vs spam.

Both methods: same GPT-2 initialization, sample order, FP32, batch 8, five epochs.

LoRA rank 4 / alpha 8; compare with full fine-tuning and pretrained two-label scoring.

Equal two-rate search (0.001 / 0.0001) per method; select by validation NLL before testing.

Rules: final five-epoch checkpoint vs first epoch with training label NLL <= 0.05.

Stopping and final checkpoints share one trajectory; no test-based checkpoint selection.

Always-ham accuracy baseline: 87.94%; spam F1: 0%.

Read spam F1 and recall alongside accuracy because the dataset is imbalanced.

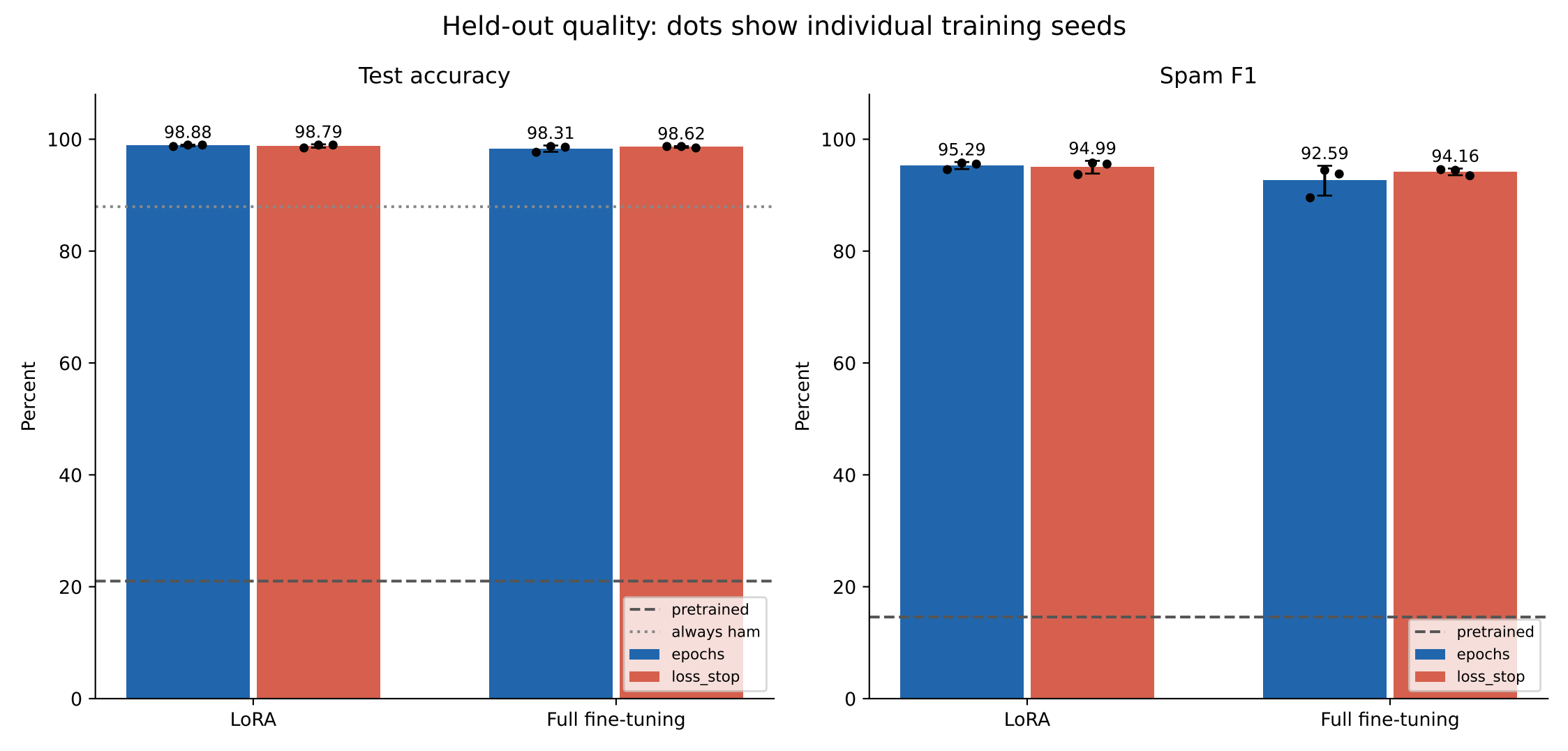
Error bars: sample SD over three training seeds, all using the same fixed test set.

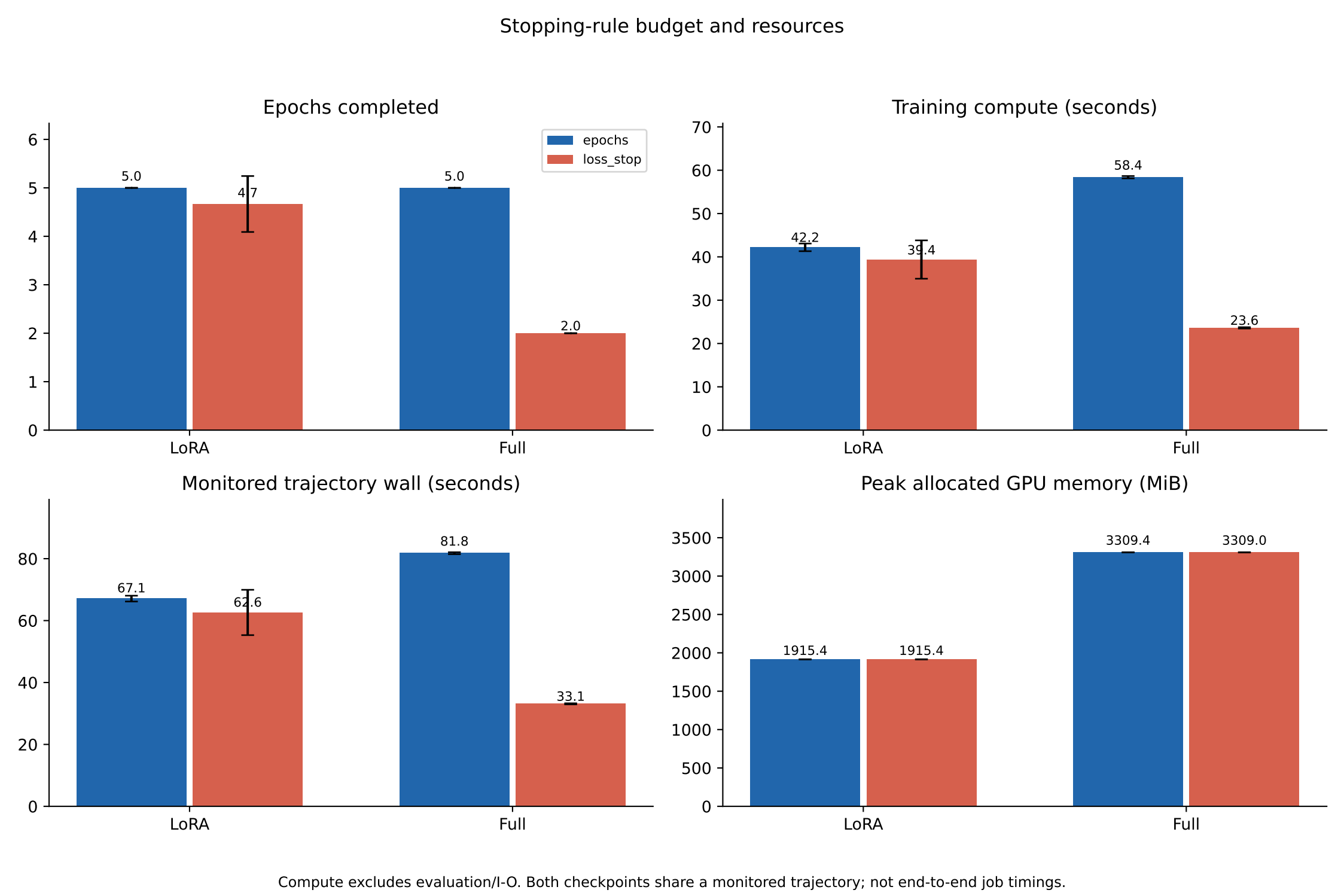
Scope: one older public dataset, limited training subset; not a production SMS filter.

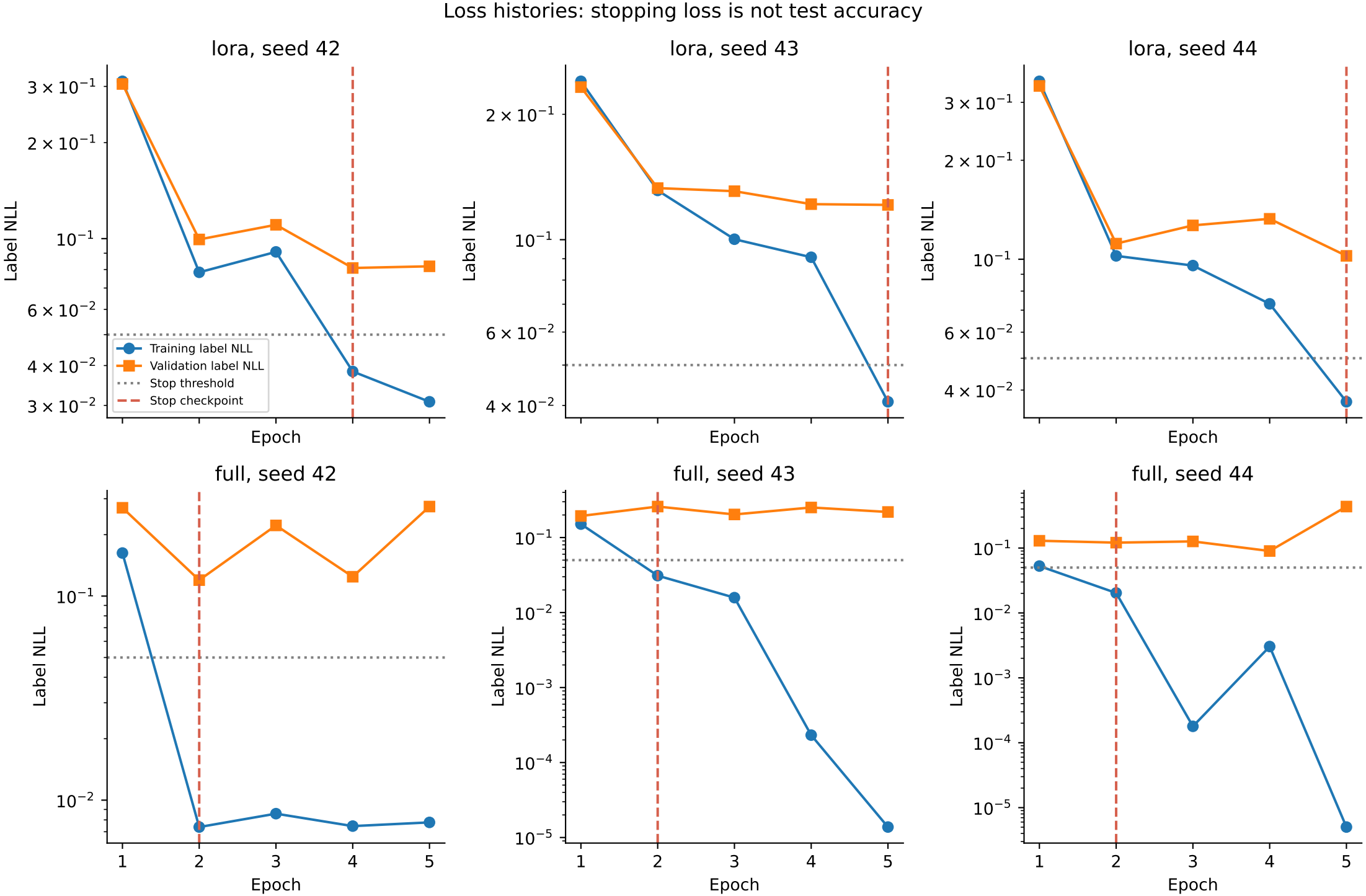
Source: UCI SMS Spam Collection, Almeida and Hidalgo (2011).

https://archive.ics.uci.edu/dataset/228/sms+spam+collection

Raw histories, every prediction, checksums and per-seed metrics accompany this PDF.


Held-out quality: dots show individual training seeds
Test accuracy
Spam F1
Percent
0
20
40
60
80
100
98.88
98.79
98.31
98.62
95.29
94.99
92.59
94.16
LoRA
Full fine-tuning
pretrained
always ham
epochs
loss_stop


Stopping-rule budget and resources
Epochs completed
epochs
loss_stop
5.0
4.7
5.0
2.0
LoRA
Full
Training compute (seconds)
42.2
39.4
58.4
23.6
LoRA
Full
Monitored trajectory wall (seconds)
67.1
62.6
81.8
33.1
LoRA
Full
Peak allocated GPU memory (MiB)
1915.4
1915.4
3309.4
3309.0
LoRA
Full
Compute excludes evaluation/I-O. Both checkpoints share a monitored trajectory; not end-to-end job timings.


Loss histories: stopping loss is not test accuracy
lora, seed 42
lora, seed 43
lora, seed 44
full, seed 42
full, seed 43
full, seed 44
Label NLL
Epoch
Training label NLL
Validation label NLL
Stop threshold
Stop checkpoint

Interpretation and limits

lora: fixed accuracy 98.88%; loss-stop accuracy 98.79%. Fixed spam F1 95.29%; loss-stop F1 94.99%.

full: fixed accuracy 98.31%; loss-stop accuracy 98.62%. Fixed spam F1 92.59%; loss-stop F1 94.16%.

Do not assume the sports-task LoRA advantage transfers. This experiment measures a different task with the same adaptation mechanism.

One-label classification and multi-token generated answers have different loss objectives. A threshold of 0.05 does not represent equal task mastery across them.

Small mean differences across three seeds do not prove equivalence or a universal winner. Training seeds do not constitute independent test datasets.

Prediction chooses between two label logits. The pretrained reference has this same restriction; no trained classifier head is added.

Training compute excludes loss checks, validation, audits, I/O and transfers. Trajectory wall time includes checks/checkpoint writes but excludes setup and final testing. Both rules share monitoring overhead.

Template grouping is heuristic; semantically similar messages may remain. The public corpus may overlap unknown pretraining data. Current SMS deployment performance is unmeasured.

Acceptance checks: identical initial weights and paired epoch sample orders; same GPU; unchanged frozen LoRA weights; matching data/source hashes; all test predictions saved.

See RESULTS.md and METRICS.csv for exact per-seed values, LR selection, wall times, confusion counts and reproduction steps.